Synchronization

Synchronization in C

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Semaphore implementations

- Mutexes in C are
  - Represented by object of type **mtx_t**
  - Defined in threads.h, i.e., insert
    - #include <threads.h>

- See documentation for
  - Atomic operation and fences (barriers) in C

Barriers are introduced in unit 06 section 07

Mutual exclusion

For more operations see the reference documentation

Type	Meaning
int mtx_init(mtx_t *mtx, int muxtype);	Create a mutex (mtx) with some properties (muxtype).
void mtx_destroy(mtx *mtx);	Destrpy the mutex pointed by mtx.
int mtx_lock(mtx_t *mtx);	Blocks the calling thread until it obtain the mutex referenced by mtx.
int mtx_trylock(mtx_t *mtx);	Try to obtain the mutex referenced by mtx but it does not block the thread.
int mtx_timedlock(mtx_t *mtx, cont struct timespec *ts);	Try to obtain the mutex referenced by mtx but it blocks the thread only for a specific time.
int mtx_unlock(mtx_x *mtx);	Releases the mutex referred by mtx.